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Computer Network Security Based On Support Vector Machine Approach

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Abstract: At present, incidents of computer networks attack are significantly increasing therefore the effective Intrusion Detection Systems (IDS) are essential for information systems security. In order for the IDS to be effective, they have the capability of detecting new arrival attacks. Additionally, the correct detection rate should also be at high level whereas the low false positive detection rate is preferred. This paper proposes the new intrusion detection technique by using hybrid methods of unsupervised/supervised learning scheme. The proposed technique integrates the Principal Component Analysis (PCA) with the Support Vector Machine (SVM). The PCA is applied to reduce high dimensional data vectors and distance between vectors including its projection onto the subspace. SVM is then used to classify different groups of data, Normal and Anomalous. The results show that the proposed technique can improve the performance of anomaly intrusion detection, the intrusion detection rate and generate fewer false alarms.

Keywords: Principal Component Analysis, Intrusion Detection System, Network Security.

1. INTRODUCTION

Nowadays, computer network security becomes a critical issue. One of most network security attacks concerned is the intrusion on computer networking systems. It is thus important to develop mechanisms to against intrusions. Basically, computer networking systems are vulnerable from both non-authorized users (outsider's attacks) as well as from authorized users who abuse their privileges (insider attacks). Intrusion Detection is defined as "the process of identifying that an intrusion has been attempted, is occurring, or has occurred" [1].

Intrusion detection is the process of monitoring the events occurring in a computer system or network and analyzing them for signs of intrusions. On the other hand, intrusion is defined as attempts to compromise the confidentiality, integrity, availability, or to bypass the security mechanisms of a computer or network. An Intrusion Detection System is a program that analyzes what happens or has happened during an execution and tries to find indications that the computer has been misused. An Intrusion Detection System does not eliminate the use of preventive mechanism but it works as the last defensive mechanism in securing the system [2].

Many approaches have been proposed by using various techniques such as statistical [3], machine learning [4], data mining [5] and immunological inspired techniques [6]. However, we can classify various techniques as 2 groups. First, it is known as "Anomaly Intrusion Detection System" based on the profiles of normal behaviors of users or applications. The anomaly IDS analyzes behavior of incoming traffics to determine whether the system is being used in a different manner [7]. The second one is called Misuse Intrusion Detection System which collects attack signatures, compares a behavior with these attack signatures, and signals intrusion when there is a match.

In this paper we present the new intrusion detection technique that applied hybrid methods. Principal Component Analysis (PCA) is applied to reduce the high dimensional data vectors and distance between a vector and its projection onto the subspace reduced. The standard Support Vector Machine (SVM) is a classifier that finds a maximal margin separating two classes of data for Intrusion Detection System [8].

This paper is organized as follows Section 2 discussed the Related Work such as An Introduction to Principal Component Analysis An explanation on Support Vector Machines Section 3 provided Experimental Design and Setup. And Experimental Results are shown in Section 4. Section 5 ends the paper with a Conclusion and some discussions.

2. RELATED WORK

In the classification problem, the number of features can be quite large, many of which can be irrelevant or redundant. Since the amount of audit data that an IDS needs to examine is very large even for a small network, classification by hand is impossible. Feature reduction and feature selection improve classification by searching for the subset of features, which best classifies the training data. There are some important features that an Intrusion Detection System should possess /include. (refer in Srilatha et al. [9]).

Most intrusion occurs via network using the network protocols to attack their targets. Twycross [10] proposed a new paradigm in immunology, Danger Theory, to be applied in developing an intrusion detection system. Alves et al. [11] present a classification-rule discovery algorithm by integrating Artificial Immune Systems (AIS) and fuzzy systems. For example, during a certain intrusion, a hacker follows fixed steps to achieve his intention, by setting up a connection between a source IP address to a target IP, and then sends data to attack the target [6]. Generally, there are four categories of

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attacks [10]. They are: 1) DoS (Denial-of-Service), for example ping-of-death, teardrop, smurf, SYN flood, and the like. 2) R2L : unauthorized access from a remote machine, for example guessing password, 3) U2R : unauthorized access to local super user (root) privileges, for example, various "buffer overflow" attacks, 4) PROBING: surveillance and other probing, for example, port-scan, ping-sweep, etc. Some of the attacks (such as DoS, and PROBING) may use hundreds of network packets or connections, while on the other hand attacks like U2R and R2L typically use only one or a few connections [11].

2.1 Feature selection with Principal Component Analysis (PCA)

Feature selection is the method that selects a subset of original attributes and reduces the feature space. When the input vectors too many features and extraneous noise it can cause errors in the classification phase. Thus, selecting relevant attributes is a critical issue for competitive classifiers and for data reduction. In this context, we evaluated the performance of the PCA as a representative method.

PCA is a linear data reduction technique, which reduces a variety of data into two dimensions. It is a way of identifying patterns in data, and expressing the data in such a way as to highlight their similarities and differences. One pattern of data can be compressed by reducing the number of dimensions without loss of information.

A. PCA using covariance

The significant features are known as Eigen profiles because they are the eigenvectors (principal components) of the set of user profiles. The projection operation characterizes a user profile by a weighted sum of the Eigen profile features, so as to detect whether a user profile is anomalous, it is sufficient to compare its weights to those of known user profiles. Figure 1 shows projection distance of the normal or abnormal state.

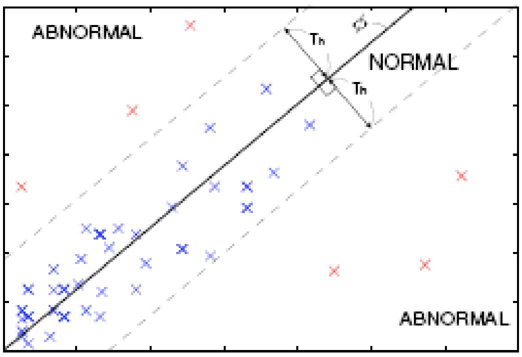

Fig. 1 Dividing projection distance into two states

Principal components are particular linear combinations of the m random variables $x_1, x_2, \ldots, x_m$ with three important properties [3]: (1) the principal components are uncorrelated, (2) the first principal component has the highest variance, the second principal component has the second highest variance, and so on, and (3) the total variation in all the principal components combined equal to the total variation in the original variables $x_1, x_2, \ldots, x_m$ The new variables with such properties are easily obtained from Eigen analysis of the covariance matrix or the correlation matrix of $x_1, x_2, \ldots, x_m$ [12].

$$X_{n\times m} = \begin{bmatrix} x_{11} & \cdots & x_{1m} \\ x_{21} & \cdots & x_{2m} \\ \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nm} \end{bmatrix} = [x_1, \ldots, x_n] \tag{1}$$

In many data sets, the first several principal components contribute most of the variance in the original data set, so that the rest can be disregarded with minimal loss of the variance for dimension reduction of the data [12, 13]. The transformation works as follows:

Given the KDDCup99 data, each data has 41 features represented by $x_1, x_2, \ldots, x_{41}$, where each observation is represented by a vector of length m, the data set is represented by a matrix X_{nxm}

The average observation is defined as:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{2}$$

The deviation from the average is defined as:

$$\Phi_i = x_i - \mu \tag{3}$$

The sample covariance matrix of the data set is defined as:

$$C = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)(x_i - \mu)^T = \frac{1}{n}AA^T \tag{4}$$

where $A = [\Phi_1, \Phi_2, \ldots, \Phi_n]$.

To apply PCA, the eigenvalues and corresponding eigenvectors of the sample covariance matrix C are usually computed by the Singular Value Decomposition (SVD). Suppose $(\lambda_1, u_1), (\lambda_2, u_2), \ldots, (\lambda_m, u_m)$ are m eigenvalue-eigenvector pairs of the sample covariance matrix C. The dimensionality of the subspace k can be determined by [12].

$$\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{m}\lambda_i} \geq \alpha \tag{5}$$

Where α is the ratio of variation in the subspace to the total variation in the original space. We form a m x k matrix U whose columns consist of the k eigenvectors. The representation of the data by principal components consists of projecting the data onto the k-dimensional subspace according to the following rules [13].

$$y_i = U^T(x_i - \mu) = U^T \Phi_i \tag{6}$$

B. PCA using Singular Value Decomposition (SVD)

Another algebraic solution for PCA and in the process, find that PCA is closely related to Singular Value Decomposition (SVD).

Definition 1. Let A be a general real *M x N* matrix. The Singular Value Decomposition (SVD) of A is the factorization.

$$A = U \sum V^T \tag{7}$$

Where U is a column-orthonormal N x r matrix, r is the rank of the matrix A, $\sum$ is a diagonal r x r matrix of the eigenvalues λ_i of A, where $\lambda_1 \geq \ldots \geq \lambda_r \geq 0$ and V is a column-orthonormal M x r matrix [14].

The eigenvalues and the corresponding eigenvectors are sorted in non-increasing order. V is called the right eigenvector matrix, and U the left eigenvector matrix. Note that the eigenvectors obtained by applying SVD to covariance matrix are the same as the principal components.

2.2 Support Vector Machine (SVM)

Support Vector Machines have been proposed as a novel technique for intrusion detection. SVM maps input (real-valued) feature vectors into higher dimensional feature space through some nonlinear mapping. SVMs are powerful tools for providing solutions to classification problems. These are developed on the Principle of Structural Risk Minimization. Structural Risk Minimization seeks to find a hypothesis for which one can find the lowest probability of error. The Structural Risk Minimization can be achieved by finding the hyper plane with maximum separable margin for the data [15].

Intrusion Detection System can automatically separate data into normal or anomalous distributions. The Intrusion Detection model based on SVM (Support Vector Machine) is mainly classified into three types. The first type [16] divides data into normal data and attack data using characteristics of the binary classifier SVM, and the second type [17] implements an anomaly detection model using one-class SVM. Finally, the third type [18] establishes multi-class SVM at the form of combining many binary classifiers, namely, SVMs and divides data into normal data and four types of attack data [19].

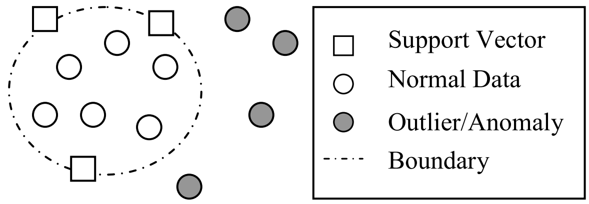

Fig. 2 Sample SVM

In Fig. 2, Normal (within bounds) and Anomalous (outliers): Those data points that define the boundary between the normal and the anomalous are called Support Vectors, thus Support Vector Machines. These “define” normalcy.

Computing the hyper plane to separate the data points, i.e. training a SVM, leads to a quadratic optimization problem. SVM uses a feature called a kernel to solve this problem. A kernel transforms linear algorithms into nonlinear ones via a map into feature spaces. SVMs classify data by using these support vectors, which are members of the set of training inputs that outline a hyper plane in feature space [20].

The kernels are very useful in finding and generalizing the principles that distinguish normal data from attacks. It is in this flexibility that the use of SVMs and kernels can greatly increase the IDS success rate at detecting new attacks.

In there is an explanation on the different kinds of learning machines as follows [21].

Two-class classifications SVM learn linear decision rules described by a weighted vector and a threshold. The idea of Structural Risk Minimization is to find a hypothesis for which one can guarantee the lowest probability of error. Geometrically, we can find two parts representing the two classes with a hyper-plane with normal distance from the origin as depicted in Figure 3.

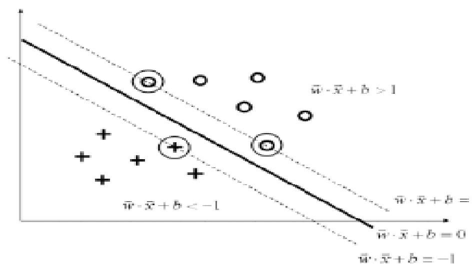

Fig. 3 Optimal separating hyper-plane is the one that separates the data.

$$D(\vec{x}) = sign\{\vec{\omega}\vec{x} + b\} = \begin{cases} +1, if\ \vec{\omega}\vec{x} + b > 0 \\ -1, \quad else \end{cases} \tag{8}$$

Supports that separating hyper-planes exist, we define the margin of a separating hyper-plane as the minimum distance between all input vectors and the hyper-plane.

Multi-class classification SVM were originally designed for two-class classification, commonly called as Binary Classification. Currently, there are two types of approaches for multi-class SVM, one-against-all approach [22] and one-against-one approach [23]. The former, is by constructing and combining several binary classifiers while the latter is by directly considering all data in one Optimization Formulation [24].

3. EXPERIMENTAL DESIGN AND SETUP

In our method, we have three steps (Figure.4). First step is for cleaning and handling missing and incomplete data. Second step is for selecting the best attribute or feature selection step using PCA. And the last step is for classifying different groups of data using SVM.

Normalization step consist of two steps.

The first step involved mapping symbolic-valued attributes to numeric-valued attributes and the second step implemented non-zero numerical features. Table 1 describes variables for Intrusion Detection Data set.

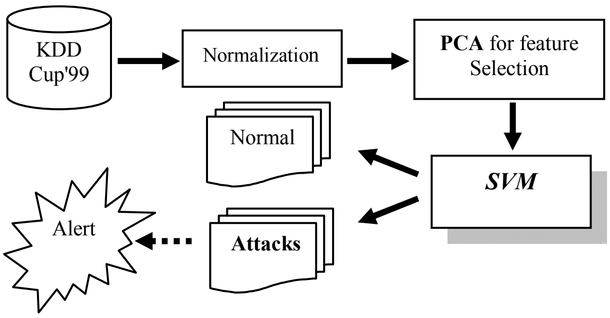

Fig. 4 The architecture of IDS Model

In this experiment, we use a Standard Dataset as the raw data used by the KDD Cup 1999 Intrusion Detection Contest [25]. This database includes a wide variety of intrusions simulated in a military network environment that is a common benchmark for evaluation of intrusion detection techniques. In general, the distribution of attacks is dominated by Probes and Denial-of-Service attacks; the most interesting and dangerous attacks, such as compromises, are grossly under-represented [26]. The data set has 41 attributes for each connection record plus one class label. There are 24 attack types, but we treat all of them as an attack group. A data set of size N is processed. The nominal attributes are converted into linear discrete values (integers). After eliminating labels, the data set is described as a matrix X, which has N rows and m=41 columns (attributes). There are md=8 discrete-value attributes and mc = 33 continuous-value attributes.

Table 1 Variables for intrusion detection data set

No.	Variable name	Type	Variable label
1	duration	continuous	A
2	protocol_type	discrete	B
3	service	discrete	C
4	flag	discrete	D
5	src_bytes	continuous	E
6	dst_bytes	continuous	F
7	land	discrete	G
8	wrong_fragment	continuous	H
9	urgent	continuous	I
10	hot	continuous	J
11	num_failed_logins	continuous	K
12	logged_in	discrete	L
13	num_compromised	continuous	M
14	root_shell	continuous	N
15	su_attempted	continuous	O
16	num_root	continuous	P
17	num_file_creations	continuous	Q
18	num_shells	continuous	R
19	num_access_files	continuous	S
20	num_outbound_cmds	continuous	T
21	is_host_login	discrete	U
22	is_guest_login	discrete	V
23	count	continuous	W
24	srv_count	continuous	X
25	serror_rate	continuous	Y
26	srv_serror_rate	continuous	Z
27	rerror_rate	continuous	AA
28	srv_rerror_rate	continuous	AB
29	same_srv_rate	continuous	AC
30	diff_srv_rate	continuous	AD
31	srv_diff_host_rate	continuous	AE
32	dst_host_count	continuous	AF
33	dst_host_srv_count	continuous	AG
34	dst_host_same_srv_rate	continuous	AH
35	dst_host_diff_srv_rate	continuous	AI
36	dst_host_same_src_port_rate	continuous	AJ
37	dst_host_srv_diff_host_rate	continuous	AK
38	dst_host_serror_rate	continuous	AL
39	dst_host_srv_serror_rate	continuous	AM
40	dst_host_rerror_rate	continuous	AN
41	dst_host_srv_rerror_rate	continuous	AO

Table 2 Dataset for attack distribution

Attack Type	Population Size
Normal	5,763
Probe	2,164
DoS	3,530
U2R	70
R2L	6,689
Summary	18,216

4. EXPERIMENTAL RESULT

4.1 Data processing

A considerable amount of data-preprocessing had to be undertaken before we could do any of our modeling experiments. It was necessary to ensure though, that the reduced dataset was as representative of the original set as possible. The test dataset that previously began with more than 300,000 records was reduced to approximately 18,216 records. Table 2 shows the dataset after balancing the different categories for attack distribution and modified the normal and other attack categories. Preprocessing consisted of two steps.

The first step involved mapping symbolic-valued attributes to numeric-valued attributes and the second step implemented non-zero numerical features. We reduce the dimensionality of this data set from 41 to 10 attributes which are duration, service, src_bytes, dst_byte, count, srv_count, serror_rate, dst_host_srv_count, dst_host_diff_srv_rate,and dst_host_same_src_port_rate

4.2 Performance measurement

Standard measures for evaluating IDSs include detection rate, false alarm rate, trade-off between detection rate and false alarm rate [27], performance (Processing Speed + Propagation + Reaction), and Fault Tolerance (resistance to attacks, recovery, and subversion). Detection Rate is computed as the ratio between the number of correctly detected attacks and the total number of attacks, while False Alarm (false positive) Rate is computed as the ratio between the numbers of normal connections that are incorrectly misclassified as attacks [28]. These are good indicators of performance, since they measure what percentage of intrusions the system is able to detect and how many incorrect classifications are made in the process.

Table 3 Experimental results of Model

Class type	# of record	Hit	Miss	Detection rate	False Alarm rate
Normal	5,763	5,751	12	99.79%	0.21%
Probe	2,164	2,159	5	99.76%	0.24%
DoS	3,530	3,253	277	92.15%	7.85%
U2R	70	67	3	95.71%	4.29%
R2L	6,689	6,521	168	97.48%	2.52%
summary	18,216	17,751	465	97.44%	2.56%

5. CONCLUSION

In conclusion we provide the new intrusion detection technique that applied hybrid algorithm. To combine between the Principal Component Analysis (PCA) and the Support Vector Machine (SVM) are the advantage of this new Intrusion Detection System. Experimental results show that the performance for Detection Rate 97.44% and False Alarm Rate 2.56% of anomaly intrusion detection. The evaluated our approach on the benchmark data from KDDCup'99 data set. Intrusion Detection Model is a composition model that needs various theories and techniques. One or two models can analysis the hardly offer satisfying results. We plan to apply other theories and techniques in intrusion detection in our future work.

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